

Logical Equivalence:-

$$\underline{P \Leftrightarrow Q} = \underline{(P \wedge Q) \vee (\sim P \wedge \sim Q)}$$

| P | Q | $P \Leftrightarrow Q$ | $(P \wedge Q)$ | $(\sim P \wedge \sim Q)$ |
|---|---|-----------------------|----------------|--------------------------|
| T | T | T                     | T              | F                        |
| T | F | F                     | F              | F                        |
| F | T | F                     | F              | F                        |
| F | F | T                     | F              | T                        |

$$\rightarrow P \Rightarrow Q = \underline{(\sim P) \Rightarrow (\sim Q)}$$

$$(\sim Q) \Rightarrow (\sim P)$$

De Morgan's law

$$1. \sim (P \wedge Q) = \sim P \vee \sim Q$$

$$2. \sim (P \vee Q) = \sim P \wedge \sim Q$$

Ex. 40  $\sim (\underline{P \vee Q}) = (\underline{\sim P}) \wedge (\underline{\sim Q})$

| P | Q | $\sim P$ | $\sim Q$ | $P \vee Q$ | $\sim (P \vee Q)$ | $(\sim P) \wedge (\sim Q)$ |
|---|---|----------|----------|------------|-------------------|----------------------------|
| T | T | F        | F        | T          | F                 | F                          |
| T | F | F        | T        | T          | F                 | F                          |
| F | T | T        | F        | T          | F                 | F                          |
| F | F | T        | T        | F          | T                 | T                          |